SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 0603 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 10%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Multiple Choice Questions(MCQ)

Code of Conduct:

I X. Joshua – (192312188) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

X. Joshua

1. b) To update prior beliefs using new evidence
2. b) Features are conditionally independent given the class
3. c) Maximize likelihood of observed data
4. b) Prefers simplest model that explains data well
5. b) By averaging predictions over all hypotheses
6. a) Selects hypothesis randomly from consistent set
7. c) Features being conditionally independent
8. a) Nodes represent variables and edges represent dependencies
9. b) Computes expected values of hidden variables
10. c) Model parameters maximizing likelihood
11. b) To estimate underlying probability distributions
12. b) Number of samples needed for learning
13. b) Restricts possible models considered
14. b) Allows continuous and complex models
15. b) Maximum number of mistakes made
16. b) Initial belief before observing data
17. b) Using prior and likelihood with evidence
18. b) Probability of data given hypothesis
19. b) Classification problems with high-dimensional data
20. b) Model captures noise instead of pattern
21. b) Fails to learn patterns from data
22. b) When parameters stabilize across iterations
23. c) Both are combined to update belief
24. b) To find hypothesis consistent with data
25. b) Exploring possible hypotheses to find best fit
26. a) It acts as a scaling factor
27. b) To avoid numerical underflow
28. b) Probability of an event given another event has occurred
29. b) Independence assumption violation
30. b) Conditional independence
31. b) Guides learning before observing data
32. b) Likelihood estimation becomes more accurate
33. b) Ignores prior information
34. b) Uses both prior and likelihood

35. b) Handles zero probability problem
36. b) Assumed model for mapping inputs to outputs
37. b) Define all possible models
38. b) Performing well on unseen data
39. b) Leads to overfitting
40. b) Choosing initial parameters
41. b) Stabilization of parameters
42. b) Handling uncertainty
43. c) Measures fit of data to hypothesis
44. b) Model fits noise in training data
45. b) Model fails to capture patterns
46. b) Guides model learning
47. b) Proper regularization
48. b) Selecting relevant features
49. b) Assign labels to data
50. b) Predict continuous values